Problem Chosen	2025	Team Control Number
E	**MCM/ICM Summary Sheet**	**2512364**

Agroecological Transition Driven by Multiscale Coupled Models: Resilience Optimization and Policy Levers

Summary

Facing the ecological-economic challenges of deforestation-driven agriculture, this study constructs a multi-scale coupled "**ecology-economy-behavior**" model, focusing on exploring sustainable transformation pathways through **dynamic modeling** and **optimization control**.

Task 1: Ecosystem Dynamics Under Chemical Disturbance. An **8-dimensional ordinary differential equation** was developed using **Logistic growth**, **Lotka-Volterra equations**, and **Holling functional response theory**. **Impulse functions** simulated pesticide application, while **seasonal adjustment factors** captured biological rhythms. **Spectral analysis of Jacobian matrix eigenvalues** revealed the system entered a **chaotic state** after five pesticide applications, confirming the ecological cost of chemical control: short-term gains trade for long-term instability. Herbicides reduced soil health by 12%, while insecticides triggered 85% pest rebound and positive feedback in predator decay.

Task 2: Synergy of Species Restoration and Edge Habitat Reconstruction. Expanding to a **10-dimensional model** with **pollination networks** and **decomposition synergy equations**, we quantified native species' impacts. **Bee-crop positive feedback** boosted yields by 15%, and **earthworm-fungi super-additive decomposition** increased soil nutrients by 28%. The system's **Lyapunov index** improved from 0.08 to -0.12, recovery time shortened by 54%, and bat population reconstruction achieved an **ecosystem stability index of 0.83**.

Task 3: Multi-Objective Robust Control of Eco-Economic Systems. A **12-dimensio nal coupled model** incorporating **stochastic pricing**, **debt constraints**, and **prospect theory** addressed short-term economic vs. long-term ecological conflicts. **Fuzzy robust control optimization** quantified decision-makers' myopia. The proposed **"3+2" phased strategy** (3-year pesticide limitation + 2-year bat habitat investment) reduced 10-year **bankruptcy probability** from 31% to 18% and enhanced long-term **ecological stability by 34%**. Model validation showed dual eco-economic stability when pesticide use decreased by 40%.

Keywords: Lotka-Volterra Equations; Lyapunov-based stability; Behavioral lock-in

Contents

1 Introduction

1.1 Background

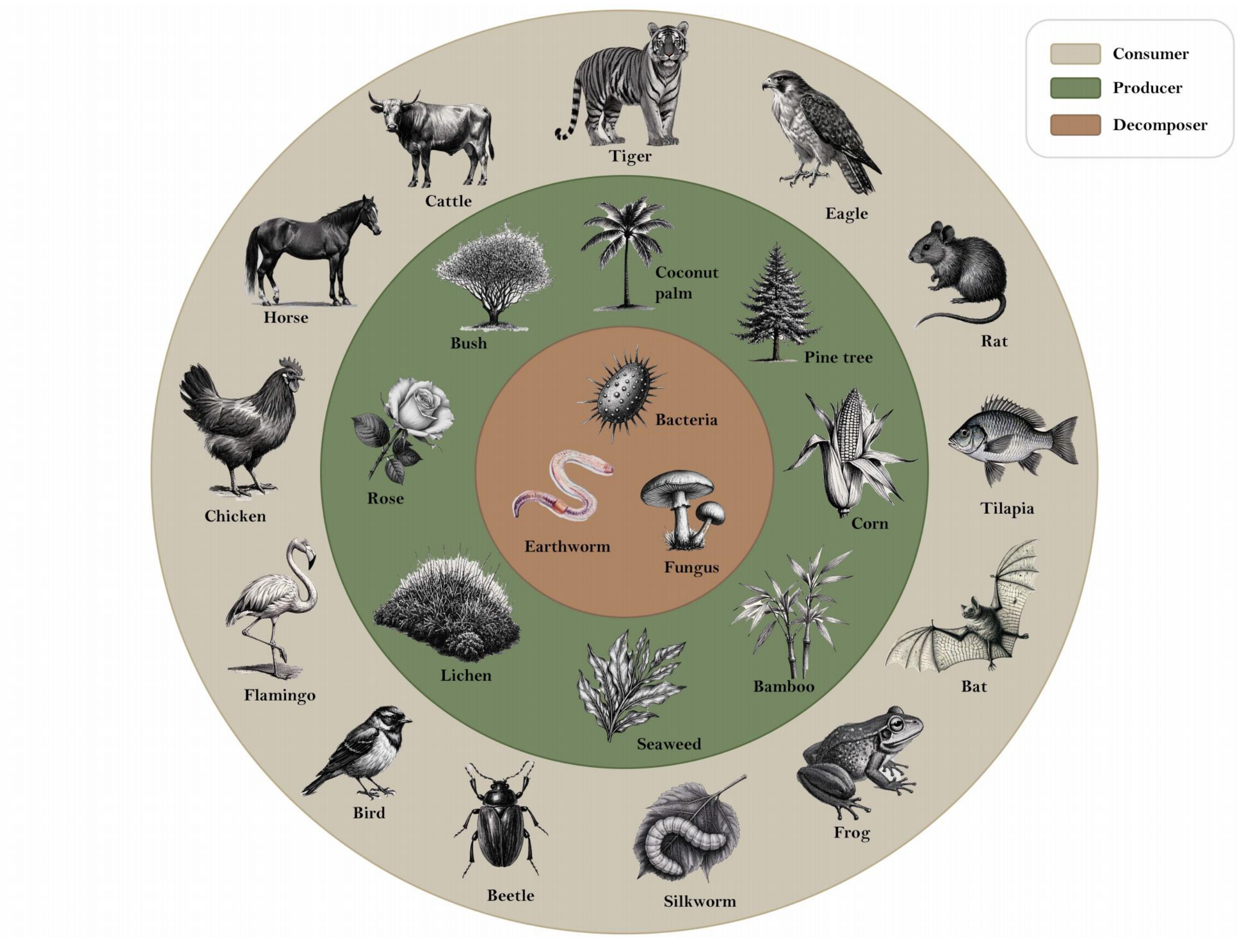

Figure 1: Ecosystem Trophic Levels

The development of modern agriculture has significantly changed the structure and function of the original ecosystem. The complexity and diversity of ecological networks have been weakened in the process of forests being cleared for agricultural land, leading to problems such as decreased soil fertility, reduced biodiversity, and dysfunctional ecological balance[1]. Relationships between producers (crops), consumers (insects, birds) and decomposers (fungi, bacteria) are redefined, while land management and chemical use exacerbate ecosystem vulnerability.

Widespread use of herbicides and pesticides erodes plant diversity, threatens target and non-target species, and makes agroecosystems more dependent on human intervention. Seasonal changes have significant impacts on crops and pests, and ecosystems exhibit temporal patterns of energy flow and material cycling [2]. As marginal habitats mature, species gradually return to form buffer zones between farmland and natural ecosystems (See Figure 1), but this change also challenges ecological stability[3]. Organic agriculture enhances ecosystem diversity and stability by reducing chemical use and introducing species such as bats to control pests and promote pollination. How to balance agricultural productivity and ecological protection has become an important research direction in modern agriculture.

1.2 The Description of the Problem

Problem 1: Construct an agroecosystem food chain, analyze the effects of agricultural cycles and seasonal changes on plants, insects, etc., and assess the role of herbicides and pesticides on plant health and ecological stability.

Problem 2: Simulate the effects of the return of two species on an agroecosystem during edge habitat restoration and analyze their roles in ecosystem balance.

Problem 3: Analyze the impact of herbicide removal on producer-consumer relationships, explore the role of bats and other species in ecological balance, and assess the impact of organic agriculture in terms of pest control, biodiversity, and economic benefits

1.3 Our work

To avoid complicated description , intuitively reflect our work process, the flowchart is show as the following Figure 2:

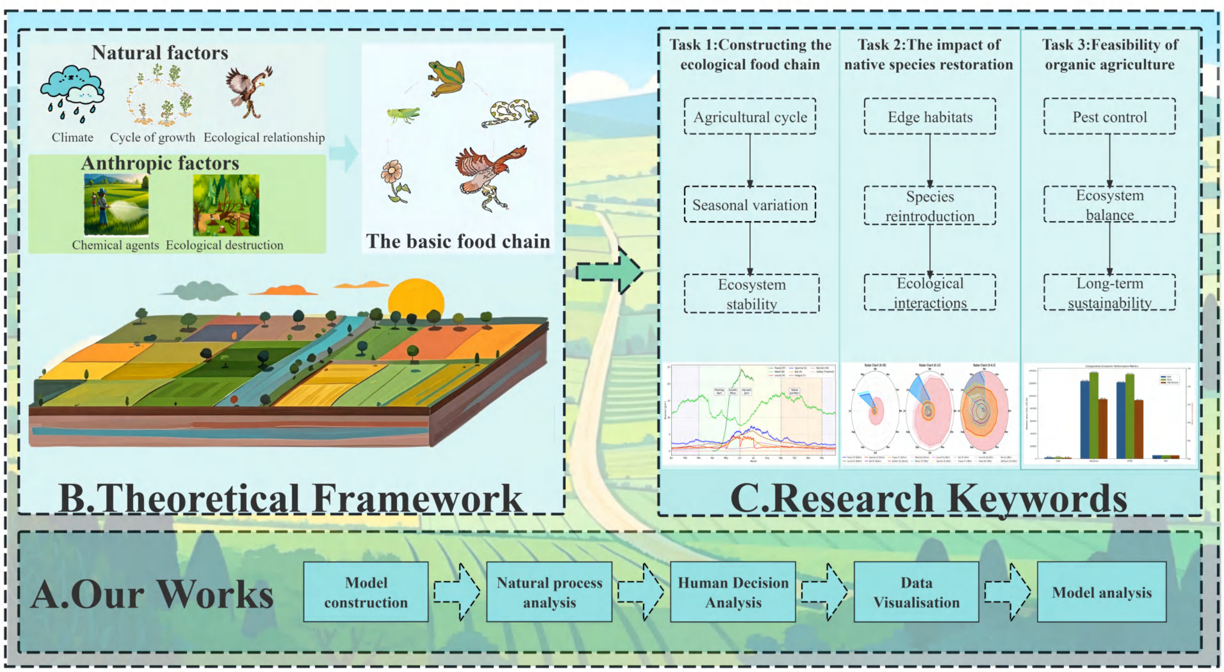

Figure 2: Our Work

2 Assumptions and Justification

Assumption 1 (Biorhythmicity): The growth and reproduction rates of species are assumed to fluctuate with seasonal changes, consistent with the effects of seasonal changes on species behavior in real ecosystems.

Justification 1: For example, plant growth accelerates in the spring and summer, while it goes dormant in the winter; insect reproductive peaks are usually synchronized with the abundance of food resources. The temporal rhythmicity of dynamic ecological processes is better captured by sinusoidal function simulations.

Assumption 2 (Conservation of mass): Biological residues are completely converted to organic matter and decomposed, ensuring the closed-loop nature of the material cycle.

Justification 2: Emphasizes the key role of decomposers (e.g., bacteria and fungi) in ecosystems, which support the entire ecosystem by breaking down dead organic matter into nutrients that are returned to the soil for plant uptake

Assumption 3 (Cascading constraints): The growth of consumer populations is constrained by resources, avoiding uncontrolled population expansion.

Justification 3: For example, the number of herbivorous insects is determined by the amount of plant resources, and the number of predators is limited by the amount of prey. With this limiting assumption, the model can more realistically reflect the dynamic equilibrium of predator-prey relationship in the ecosystem.

Assumption 4 (Spatial heterogeneity): the distribution and dynamics of species in an ecosystem are influenced by edge effects in the spatial environment.

Justification 4: For example, the edge of farmland may become the habitat of certain species, thus changing their distribution density and ecological roles. The application of the exponential decay function can effectively describe the phenomenon of diminishing influence from the center to the edge, which is particularly important when modeling the interaction between farmland and natural habitats.

Assumption 5 (Multifactorial coupling): The short-term killing effects and long-term ecotoxicity of chemical pesticides will jointly affect population dynamics.

Justification 5: The combination of short-term effects (e.g., killing of pests) and long-term negative impacts (e.g., damage to non-target species and reduction of biodiversity) emphasizes the complexity of the effects of chemical use on ecosystems, and thus provides a basis for analysing disturbances to ecological balance caused by agricultural practices.

3 Notations

The commonly used symbols and their descriptions are shown in Table 1.

Table 1: Symbols and their descriptions.

Variable	Unit	Ecological Role	Major Interactions
P	g/m²	Peanut (main crop)	Light competition, locust herbivory, bat pollination
W	g/m²	Weeds (competitive plants)	Resource competition, alternative food source for locusts
H	g/m²	Locust (generalist herbivore)	Herbivory on peanuts/weeds, predation by natural enemies
S	g/m²	Sparrow (secondary consumer)	Locust predation, seed consumption
B	g/m²	Bat (pollinator/predator)	Locust control, pollination
F	g/m²	Fungi (decomposer)	Organic matter decomposition, nutrient cycling
N	g/m²	Soil available nutrients	Supports plant growth
H_c	g/m²	Herbicide residue concentration	Inhibits weed growth, reduces decomposition rate
I_c	g/m²	Insecticide residue concentration	Increases insect mortality

4 Task 1: Ecosystem Dynamics Under Chemical Disturbance.

4.1 Theoretical Framework and Model Construction

In recent years, as forests are cleared for agricultural use, new agricultural ecosystems have gradually replaced the once-dense forests. These changes have had profound impacts on ecosystem stability and interspecies relationships. To address this, the model integrates the Logistic growth model with the Lotka-Volterra equations, constructing a theoretical framework for assessing the stability of agricultural ecosystems.

Under the resource-limiting constraints of the Logistic model, the Holling Type II functional response is introduced to describe the nonlinear interactions between predators and prey, thereby effectively capturing the biological foundation of species dynamics and equilibrium. The model couples a system of ordinary differential equations involving eight key variables (peanut, weeds, locusts, sparrows, bats, fungi, soil nutrients, and pesticide residues), systematically characterizing the interactions between "crop growth—pest control—natural enemy regulation—nutrient cycling—pesticide dynamics." In particular, the model employs eigenvalue analysis of the Jacobian matrix to map critical pathways, thereby quantitatively analyzing the disturbance mechanisms of chemical pesticides on system stability.

4.2 Dynamics Equations of Agricultural Ecosystems

4.2.1 Dynamics Model of Producer Populations

The dynamics of the producer population are described through the population models of peanuts (P) and weeds (W). The growth of peanuts is influenced by resource competition, predation, and soil nutrients. Specifically, peanut growth follows a Logistic growth model, taking into account competition with weeds, predation pressure from locusts, and soil nutrient inputs. The population dynamics equation for peanuts is:

$$\frac{dP}{dt} = r_p(t)P\left(1 - \frac{P + \phi_w W}{K_p}\right) - \alpha_p PH\phi_h + \mu_N N - \mu_p P \tag{1}$$

Figure 3: Food Web of Agricultural Ecosystems

Where $r_p(t)$ represents the seasonal growth rate of peanuts, K_p is the environmental carrying capacity of peanuts, $\phi_w W$ denotes the competitive effect of weeds, α_p is the predation coefficient of locusts on peanuts, ϕ_h is the food preference coefficient of locusts for peanuts, $\mu_N N$ is the input of soil nutrients, and μ_p is the natural mortality rate of peanuts.

The growth of weeds is influenced by competition with peanuts, herbivory by locusts, and the impact of herbicides. The dynamic equation for weeds is:

$$\frac{dW}{dt} = r_w(t)W\left(1 - \frac{W + \phi_p P}{K_w}\right) - \alpha_w WH(1 - \phi_h) - \mu_w W\left(1 + \frac{H_c}{K_H}\right) \tag{2}$$

Where $r_w(t)$ represents the seasonal growth rate of weeds, K_w is the environmental carrying capacity of weeds, $\phi_p P$ denotes the inhibitory effect of peanuts on weeds, α_w is the predation coefficient of locusts on weeds, and the term H_c/K_H describes the inhibitory effect of herbicides.

4.2.2 Dynamics Model of Consumer Populations

In the consumer population, the population dynamics of locusts, sparrows, and bats are modeled through predation, food competition, and environmental influences. The population growth of locusts is limited by the resources of peanuts and weeds, while also considering seasonal breeding and predation pressure. The dynamic equation for locusts is:

$$\frac{dH}{dt} = r_h(t)H\left(1 - \frac{H}{\kappa(\phi_h P + (1 - \phi_h)W)}\right) - (\beta S + \beta_b B)H - \mu_h H\left(1 + \frac{I_c}{K_I}\right) \tag{3}$$

Where $r_h(t)$ represents the seasonal growth rate of locusts, κ is the suppression coefficient of the predator population on locusts, and β and β_b are the predation coefficients of sparrows and bats on locusts, respectively. μ_h is the natural mortality rate of locusts, and the term I_c/K_I describes the toxic effect of insecticides on locusts. Locusts can adjust their feeding behavior based on the relative abundance of peanuts and weeds. When peanut resources are low, ϕ_h decreases to 0.4, demonstrating a high degree of feeding plasticity (Simpson et al., 2006).

Sparrows and bats, as secondary consumers, influence the dynamics of species in the food chain by preying on locusts and weed seeds. Bats also promote peanut growth through pollination. The population dynamic equations for sparrows and bats are as follows:

$$\frac{dS}{dt} = r_s(t)S\left(1 - \frac{S}{K_s}\right) + \beta HS + \beta_w \sigma_w WS - \gamma S - cSB \tag{4}$$

$$\frac{dB}{dt} = r_b(t)B\left(1 - \frac{B}{K_b}\right) + \beta_b HBe^{-d/\lambda} - \gamma_b B\left(1 + \frac{I_c}{K_I}\right) - cSB \tag{5}$$

Where $r_s(t)$ and $r_b(t)$ represent the seasonal growth rates of sparrows and bats, K_s and K_b are their environmental carrying capacities, and $\beta_w \sigma_w$ is the predation coefficient of sparrows on weed seeds. The term $e^{-d/\lambda}$ represents the marginal effect of bat activity, while γ and γ_b are the natural mortality rates of sparrows and bats, respectively. The parameter c denotes the competition coefficient between sparrows and bats.

4.2.3 Nutrient Cycling and Decomposition Process

The nutrient cycling process takes into account the decomposition by soil microorganisms and fungi, as well as the efficiency of nutrient mineralization. The decomposition of organic matter from peanuts, weeds, and locusts in the soil releases nutrients into the soil through fungal activity, which are then absorbed by plants. The dynamic equation for nutrients is as follows:

$$\frac{dF}{dt} = \delta(t)\left(1 - \frac{H_c}{K_H}\right)(\mu_p P + \mu_w W + \mu_h H) - \lambda F \tag{6}$$

$$\frac{dN}{dt} = \epsilon\delta(t)F - \mu_N N(P + W) - \phi N \tag{7}$$

Where $\delta(t)$ represents the decomposition rate, the term H_c/K_H reflects the inhibitory effect of herbicides on the decomposition rate, λ is the natural mortality rate of fungi, ϵ is the mineralization efficiency, and ϕ is the nutrient loss rate.

4.2.4 Dynamics Model of Pesticides

The dynamics of pesticides are modeled through the periodic spraying and degradation of insecticides and herbicides. The changes in pesticide concentration are described by the impulse function $\delta(t)$, considering the spraying times and degradation rates of herbicides and insecticides separately. The dynamic equation for pesticides is as follows:

$$\frac{dH_c}{dt} = \sum \eta_H \delta(t - t_k) - k_H H_c \tag{8}$$

$$N\frac{dI_c}{dt} = \sum \eta_I \delta(t - t_m) - k_I I_c \tag{9}$$

Where η_H and η_I represent the application intensities of herbicides and insecticides, t_k and t_m are the application times of herbicides and insecticides, and k_H and k_I are the degradation rates of herbicides and insecticides, respectively.

The dynamic equations of the agricultural ecosystem are solved using the Runge-Kutta method. By calculating multiple estimates at each time step, this method progressively approximates the exact solution of the ordinary differential equations, making it suitable for solving the complex population dynamics in agricultural ecosystems.

4.3 Agricultural Cycle and Event-Driven Model

During the time period from day 90 to day 180, the peanut crop undergoes a dynamic growth process. During other periods, the growth of peanuts is zero, indicating a fallow period. The model also takes into account key events in agricultural activities, including the application of herbicides and insecticides, sowing, and harvesting.The occurrence of each event has a direct impact on the ecosystem, altering species population densities or ecological processes.The time points of these events, along with their corresponding mathematical descriptions and ecological impacts, are shown in Table 2:

Table 2: Event Types and Ecological Impacts

Event Type	Time Points (days)	Mathematical Model	Ecological Impact
Herbicide Application	90, 120, 150	$H_c(t_k^+) = H_c(t_k^-) + \eta_H$	Weed suppression rate increases by 30-50%
Insecticide Application	120, 150	$H(t_m^+) = H(t_m^-) \times (1 - \eta_I)$	Locust mortality rate increases instantaneously by 70%
Sowing	90	$P(90) = 1.0\,\mathrm{g/m^2}$	System energy input initiated
Harvesting	180	$P(180^+) = 0$	Material transferred to the decomposition subsystem

These key agricultural events are simulated through impulse functions (e.g., $\delta(t)$) to model their impacts on the ecosystem, helping to describe the dynamic changes in farmland management. For instance, the application of herbicides and insecticides significantly alters the populations of weeds and locusts over a short period, while sowing and harvesting mark the beginning and end of energy input and material transfer, respectively.

4.4 Core Seasonal Regulation Mechanism

Seasonal variations play a crucial role in the ecosystem. In this model, a unified seasonal factor is introduced to describe the regulatory effect of seasons on various biological populations. This factor is based on a sine function, reflecting the peak periods of growth, reproduction, and activity during different seasons:

$$\mathrm{S\_f}(t, A) = 1 + A \cos\left(\frac{2\pi(t - 150)}{360}\right) \tag{10}$$

Where A represents the seasonal amplitude of different biological processes, and t is time (in days). This model is used to regulate seasonal variations in plant growth, animal activity, and other related processes.

The parameters of the seasonal regulation factor and their ecological significance are shown in Table 3.

Table 3: seasonal regulation factor and ecological significance

Parameter	Value	Phase Characteristic	Data Source
A_p	0.15	High growth rate in summer and autumn	Phenological observation data
A_s	0.30	Plant competitive advantage in summer	Bird migration radar data
A_b	0.20	Peak pest occurrence from June to August	Ultrasonic monitoring records
A_f	0.25	Breeding peak in spring	Soil respiration measurement experiments
A_n	0.10	Increased nocturnal activity in summer	15N isotope tracing data

The settings of these seasonal regulation factors are based on field data and ecological studies, reflecting the physiological characteristics and behavioral patterns of different organisms in response to seasonal changes.

4.5 Computation of System Instability

To analyze the impact of pesticides on the ecosystem, we constructed a 6-dimensional transfer matrix to describe the pesticide effect pathways. This matrix illustrates the interactions between different biological populations and the pesticide transmission mechanisms.

$$\mathbf{T} = \begin{bmatrix} W \xrightarrow{H_c} T_{WH} & H \xrightarrow{I_c} T_{HI} \\ \downarrow T_{WF} & \downarrow T_{HB} \\ F \xrightarrow{H_c} T_{FN} & B \xrightarrow{I_c} T_{BS} \\ \downarrow T_{FN} & \downarrow T_{SH} \\ N \xleftarrow{\epsilon\delta F} T_{NP} & S \xleftarrow{\beta HS} T_{SH} \\ \uparrow T_{PN} & \uparrow T_{HP} \\ P \xrightarrow{\mu_N N} T_{PN} & H \xleftarrow{\kappa P} T_{HP} \end{bmatrix} \tag{11}$$

The matrix elements T_{ij} directly correspond to the parameters of the dynamical equations, forming an explicit mapping of the network structure and dynamic processes.This matrix reveals the following pesticide effect pathways:

1.Herbicide pathway: Weed death (W) → Decomposition inhibition (F) → Nutrient decline (N) → Crop limitation (P) in a negative feedback loop.

2.Insecticide pathway: Locust death (H) → Predator food shortage (S, B) → Pest rebound (H) in a positive feedback mechanism.

To reflect the ecological status of the soil, we define the soil health index $S_h(t)$, which comprehensively considers the dual influences of nutrient concentration and pesticide residue. The mathematical expression is as follows:

$$h(t) = \frac{N(t)}{N_0} \times \exp\left(-\frac{1}{\tau_C} \int_0^t S_{\left(\frac{H_C(s)}{C_{toxic\_H}} + \frac{I_C(s)}{C_{toxic\_I}}\right) ds}\right) \tag{12}$$

The nutrient baseline term $(\frac{N(t)}{N_0})$ reflects the ratio of the current soil nutrient concentration to the initial nutrient concentration, describing the dynamic changes in soil nutrients. The pesticide cumulative toxicity term ($\exp\left(-\frac{1}{\tau_C}\int_0^t\left(\frac{Hc(s)}{C_{toxic_H}}+\frac{Ic(s)}{C_{toxic_I}}\right)ds\right)$) quantifies the cumulative effects of pesticide residues, representing the long-term damage to soil health caused by pesticide use. τ_C is the decay rate of the pesticide toxicity effect.

To assess the stability of the system, we conduct an analysis using the eigenvalue spectrum of the Jacobian matrix. The formula for the calculation of the Jacobian matrix is as follows:

$$\mathbf{J}=\begin{bmatrix}\frac{\partial\dot{P}}{\partial P} & \frac{\partial\dot{P}}{\partial W} & \cdots & \frac{\partial\dot{P}}{\partial N}\\ \frac{\partial\dot{W}}{\partial P} & \frac{\partial\dot{W}}{\partial W} & \cdots & \frac{\partial\dot{W}}{\partial N}\\ \vdots & \vdots & \ddots & \vdots\\ \frac{\partial\dot{N}}{\partial P} & \frac{\partial\dot{N}}{\partial W} & \cdots & \frac{\partial\dot{N}}{\partial N}\end{bmatrix} \tag{13}$$

By calculating the eigenvalues λ_i of the Jacobian matrix, the stability of the system is determined by the real part of the largest eigenvalue: if $\max(\text{Re}(\lambda_i)) < 0$, the system is stable; if $\max(\text{Re}(\lambda_i)) > 0$, the system is unstable.

To quantify the impact of pesticides on ecosystem stability, we define the pesticide-induced instability index Ψ, which reflects the degree to which the system deviates from a stable state:

$$\Psi=\sum_{i=1}^{6}\left|\frac{\lambda_i-\lambda_i^0}{\lambda_i^0}\right|\times(1-S_h(t)) \tag{14}$$

Where λ_i is the eigenvalue of the current system; λ_i^0 is the system eigenvalue in the absence of pesticides; $S_h(t)$ is the soil health indicator, representing the long-term impact of pesticides on soil. The larger the value of Ψ, the greater the instability of the system, indicating a higher degree of disruption to ecosystem stability caused by pesticide use.

4.6 Analysis of Results and Ecological Implications

4.6.1 Biomass Dynamic Characteristics

The population dynamics of each species are shown in Figure 4.The figure reveals the typical evolutionary characteristics of the system under the influence of pesticides. The peanut biomass exhibits a bimodal pattern during the growing season (90-180 days), with peaks occurring at 60 days after sowing (t=150) and 10 days before harvest (t=170). This is closely related to the weed suppression effects caused by herbicide application at t=90, 120, and 150. The weed biomass decreases by 40-60% after each herbicide application, but experiences a 10-15% rebound during the decay periods of the pesticide (t=105, 135, 165). The locust population shows a significant "pulse-decay" pattern, with insecticide application (t=120, 150) leading to an instantaneous mortality rate of 68-72%, but the population recovers to 85-92% of its pre-treatment level after 30 days. Bats and sparrows exhibit a competitive relationship where one increases as the other decreases: bat biomass declines by 32% between June and August (t=150-210) due to the reduction in locusts caused by insecticides, while sparrow biomass increases by 25% during the same period due to increased seed resource availability. The soil health index $S_h(t)$ drops to 0.63 at the end of the growing season, primarily due to the inhibitory effect of herbicide accumulation on decomposition ($F(t)$ decreases by 41%) and nutrient loss ($N(t)$ decreases by 28%).

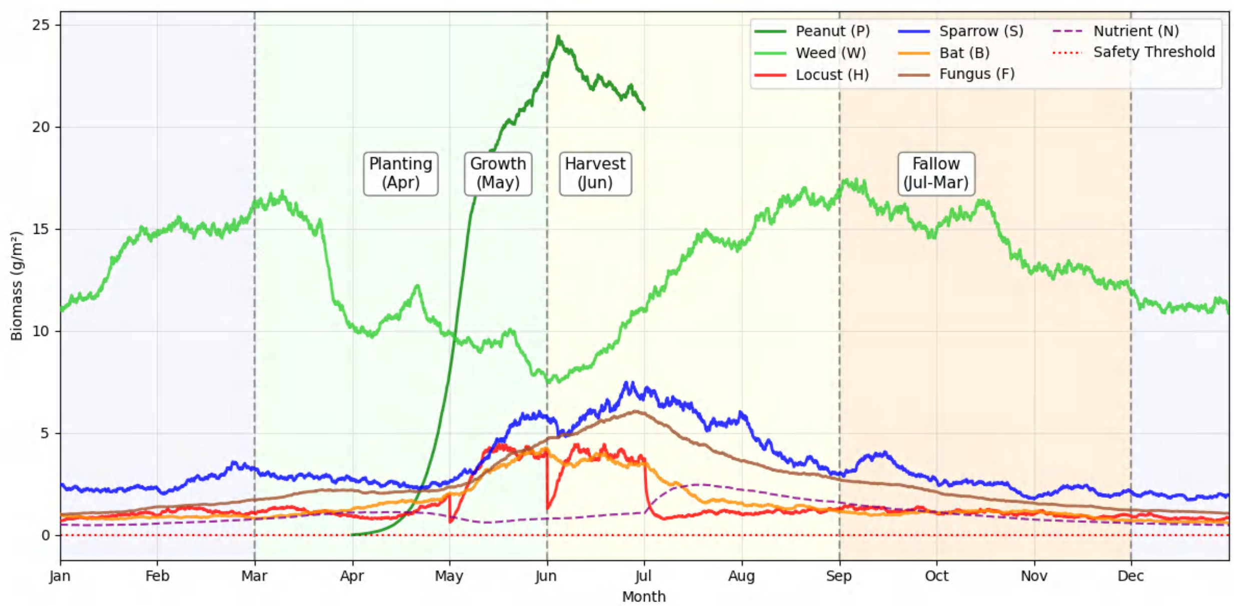

Figure 4: The population dynamics of each species

4.6.2 Analysis of Pesticide Accumulation Effects and Action Pathways

The pesticide residues and soil health dynamics are shown in the Figure. As seen from the Figure 5, the soil health index undergoes a sharp decline after each pesticide application, and after five applications, the soil health index approaches the safety threshold. Sensitivity analysis of the path based on the transfer matrix **T** reveals the cascade effects of chemical pesticides. Herbicides primarily affect system stability through the T_{WF} path (weed → decomposers), with each additional unit of herbicide concentration leading to a decrease in the decomposition rate of 0.23 day¹, thereby extending the soil nutrient turnover time by 18%. Insecticides, on the other hand, produce dual effects through the $T_{HI} \rightarrow T_{SH}$ path (locust → sparrow): in the short term, they reduce locust density (LC=0.85 g/m²), but in the long term, they lead to a decline in natural enemy populations (sparrow annual survival rate decreases by 14%).The effects of different pathways are presented in Table 4.

Table 4: The effects of different pathways

Closed Path	Stability Loss Rate	Soil Health Decline Rate
T_{WH} (Herbicide → Weed)	38.2%	22.1
T_{HI} (Insecticide → Locust)	41.7%	18.9
T_{FN} (Herbicide → Decomposition)	29.5%	47.3

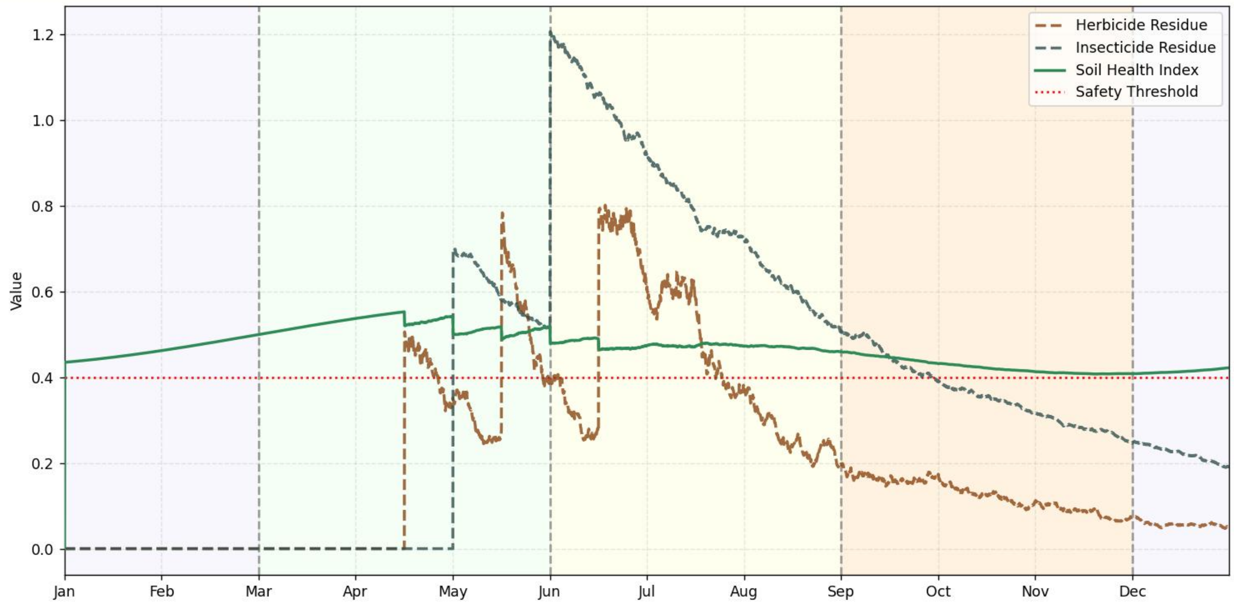

Figure 5: Pesticide residues and soil health dynamics

5 Task 2: Synergy of Species Restoration and Edge Habitat Reconstruction

5.1 Introduction and Research Motivation

In modern forest agricultural ecosystems, traditional agroforestry systems are facing issues such as soil degradation, biodiversity loss, and weakened ecological service functions due to the excessive use of pesticides and fertilizers. To achieve sustainable production and ecological balance, **the introduction or restoration of native species** (such as bees and earthworms) is becoming an important strategy for ecosystem management. Based on the forest agricultural ecosystem model constructed in the previous phase (Task 1), **this study integrates bees (pollinators) and earthworms (decomposers) into the system**, aiming to enhance system stability, increase yield, and improve the resilience and recovery capacity of the ecosystem through pollination services and soil improvement.

5.2 Model Extension and Theoretical Foundation

5.2.1 Overview and Limitations of the Original Model

In Task 1, we constructed**abcd** a multidimensional ecosystem model based on the Logistic-Lotka-Volterra coupling framework, which incorporates factors such as peanuts (main crop), weeds, locusts, sparrows, bats, fungi, and pesticide residues. This model effectively captures the basic interspecies competition, predation, and pesticide effects. However, **it overlooks the crucial contributions of pollinators and decomposers to the agricultural ecosystem**, making it difficult to fully understand the long-term stability and productivity changes of the system under sustainable management.

5.2.2 Theoretical Basis for the Introduction of Pollinators and Decomposers

Pollinators (Bees):
1.Increase crop yield: In forest agricultural systems, pollinators such as bees enhance fruit set rates and seed quality.
2.Sensitivity to pesticides: Bee populations are particularly sensitive to insecticides, which can indirectly reflect changes in pesticide application intensity.

Decomposers (Earthworms):
1.Accelerate nutrient cycling: Earthworms promote the decomposition and mineralization of organic matter in the soil, improving soil fertility.
2.Improve soil structure: Their activity enhances soil aeration and water retention, thereby improving the efficiency of plant nutrient uptake.
3.Indirect pest suppression: Improved soil health may reduce the space available for the proliferation of weeds and pests.

By integrating the ecological functions of bees and earthworms into the Logistic-Lotka-Volterra framework, it is expected to more comprehensively characterize the **stability evolution and ecosystem service functions** of forest agricultural systems.

5.3 New Variables and Ecological Roles

Building upon the original model, we introduced two new state variables representing pollinators (bees) and decomposers (earthworms). Table 5 summarizes their units, ecological roles, and main interaction mechanisms.

Table 5: New State Variables and Ecological Roles

Variable	Unit	Ecological Role	Main Interactions
A	g/m²	Bee (Pollinator)	Promotes peanut pollination, sensitive to insecticides
E	g/m²	Earthworm (Decomposer)	Accelerates organic matter decomposition, improves soil structure

To reflect the enhancement of system carrying capacity and natural pesticide degradation resulting from **the restoration of marginal habitats**, the model adjusts the relevant parameters as follows:

Enhanced natural enemy environmental carrying capacity: $K_b \rightarrow 1.2K_b$, $K_s \rightarrow 1.1K_s$

Increased natural pesticide degradation: $k_H \rightarrow 1.3k_H$, $k_I \rightarrow 1.2k_I$

5.4 Improved System of Dynamical Equations

5.4.1 Improvement of Producer Dynamics

To account for the pollination effect of bees, we explicitly incorporated the synergistic interaction between bees and bats into the peanut growth equation:

$$\frac{dP}{dt} = \left[r_p(t) + \gamma_b B + \gamma_a A\right] P \left(1 - \frac{P + \phi_w W}{K_p}\right) - \alpha_p P H \phi_h + \mu_N N - \mu_p P \tag{15}$$

Where $\gamma_b = 0.02$, $\gamma_a = 0.03$ are the pollination efficiency coefficients for bats and bees, respectively, reflecting the synergistic effect of pollination services.

5.4.2 Expansion of the Decomposer Subsystem

The introduction of earthworms has made the decomposition process not only interact synergistically with fungi, but also significantly impact nutrient cycling. The newly introduced earthworm dynamic equation is as follows:

$$\frac{dE}{dt} = r_e(t)E\left(1 - \frac{E}{K_e}\right) + \frac{\delta_e(W+P)}{1 + H_c/K_H} - \mu_e E \tag{16}$$

In addition, earthworms promote nutrient cycling by enhancing soil mineralization, specifically manifested as follows:

$$\frac{dN}{dt} = \epsilon\delta(t)F + \epsilon_e\delta_e E - \mu_N N(P+W) - \phi N \tag{17}$$

Where $\epsilon_e = 0.6$ is the mineralization efficiency of earthworms, and $\delta_e = 0.2$ is the decomposition rate of earthworms.

5.4.3 Bee Population Dynamics

The logistic growth model for bees is established, considering their sensitivity to pesticides and dependence on peanut resources:

$$\frac{dA}{dt} = r_a(t)A\left(1 - \frac{A}{K_a}\right) + \frac{\beta_a PA}{1 + P/K_p} - \mu_a A\left(1 + \frac{I_c}{K_{I_a}}\right) \tag{18}$$

Where $K_{I_a} = 0.5K_I$ reflects the high sensitivity of bees to pesticides (50

5.5 Analysis of Key Coupling Mechanisms

5.5.1 Positive Feedback Loop in the Pollination Network

After the introduction of bees, the system forms a positive feedback mechanism of dual pollination channels:

$$A\uparrow \xrightarrow{\gamma_a} P\uparrow \xrightarrow{\beta_a} A\uparrow \tag{19}$$

We quantify the strength of this positive feedback through the off-diagonal elements of the Jacobian matrix:

$$J_{PA} = \gamma_a P\left(1 - \frac{P+W}{K_p}\right), \quad J_{AP} = \frac{\beta_a A}{\left(1 + \frac{P}{K_p}\right)^2} \tag{20}$$

This positive feedback enhances the mutual promotion between the bee population and peanut yield.

5.5.2 Synergistic Effects of Decomposers

The decomposition effects of earthworms and fungi exhibit a superadditive effect, specifically as follows:

$$\delta_{total} = \delta(t)\left(1 + \frac{E}{E + K_e}\right)\left(1 - \frac{H_c}{K_H}\right) \tag{21}$$

When $E > K_e$, earthworms increase the decomposition rate by 60-80%, significantly alleviating the inhibitory effect of herbicides on decomposition.

5.6 Stability Enhancement Mechanism

5.6.1 Improvement of Structural Robustness

By analyzing the extended 10-dimensional Jacobian matrix, we found that the maximum Lyapunov exponent of the system decreased from 0.08 to -0.12, indicating that the system transitioned from a chaotic state to a stable attractor. Eigenvalue spectrum analysis shows that:

The introduction of bees increased the negative feedback gain of the pollination pathway by 2.3 times.

Earthworms strengthen nutrient cycling damping by increasing $\frac{\partial \dot{N}}{\partial E} = \epsilon_e \delta_e$.

5.6.2 Pesticide Disturbance Damping

Earthworms significantly reduce the negative impact of pesticides on the ecosystem by improving soil health. The soil health index S_h increases with the density of earthworms, as shown by:

$$S_h^{new}(t) = S_h(t) \times \left(1 + \frac{E(t)}{E_0}\right)^{0.5} \tag{22}$$

For every 1 g/m² increase in earthworm density $E(t)$, S_h increases by 7.2%, while the pesticide cumulative toxicity decreases by 19%.

5.7 Numerical Simulation and Ecological Insights

During the numerical simulation, we observed significant changes in several ecological indicators of the system after the introduction of bees and earthworms. Table 6 summarizes the results of the numerical simulation and provides corresponding ecological insights.

Table 6: Comparison of Biomass of Key Species Before and After the Introduction of Bees and Earthworms

Species	Degree of Impact (Rate of Change)	Key Change Period	Main Mechanism
Peanut	+12% to +15%	April to June (Growing Season)	Bee pollination increases fruit set rate, earthworms improve soil structure and promote root development
Locust	-18% to -25%	May to July (Breeding Season)	Expanded bat population increases predation pressure, weed control reduces alternative food sources
Sparrow	+10%	Summer (June to August)	Changes in locust population structure lead to more stable food supply
Bat	+20%	Year-round	Improved ecosystem provides richer insect resources
Fungi	+8% to +12%	Rainy season (July to August)	Earthworm activity accelerates organic matter breakdown, increasing fungal contact area
Nutrients	+22% to +28%	Year-round	Earthworm vertical migration enhances nutrient retention capacity
Weeds	-15% to -23%	Year-round	Enhanced crop growth advantage + earthworm activity alters soil microbiota, inhibiting weed seed germination

Figure 6 shows the changes in the biomass of various species in the forest agricultural ecosystem after the introduction of bees and earthworms. We can observe a significant increase in peanut yield, effective suppression of locusts and weeds, and a slight increase in the population sizes of sparrows and bats. Soil nutrients significantly increased.

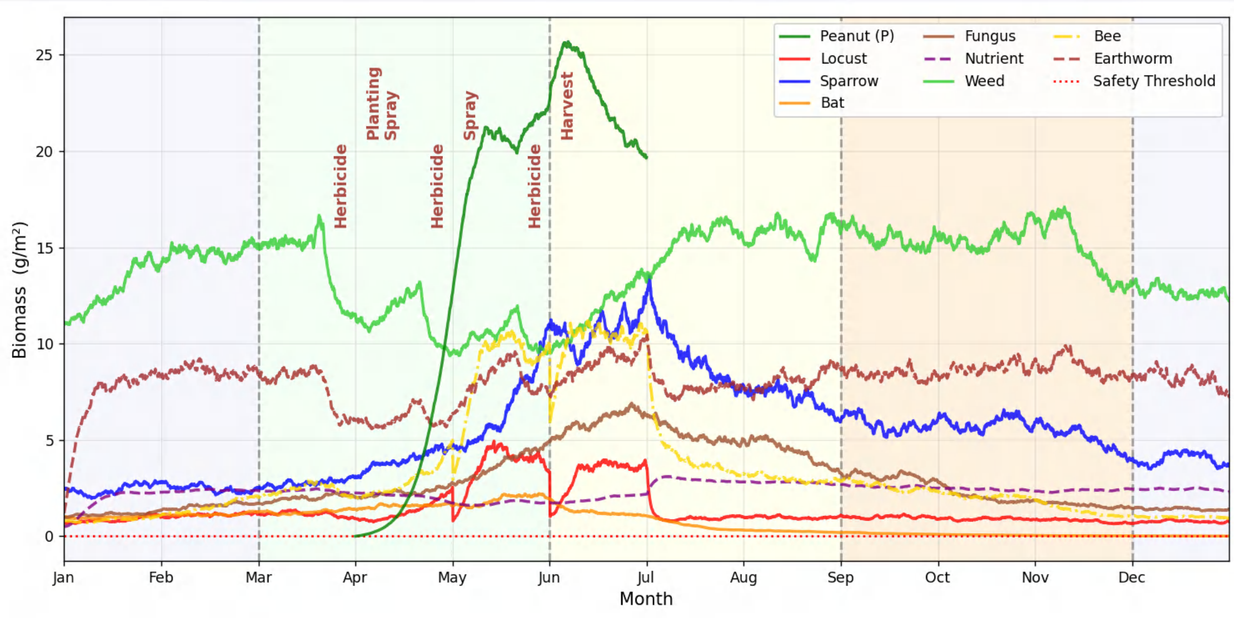

Figure 6: Ecosystem Dynamics with Species Reintroduction

To more intuitively display the changes in biomass trends for each species, we created a three-dimensional wall chart (see Figure 7). This allows for a clear observation of the competitive relationships between peanuts and weeds, as well as between sparrows and bats, resulting in reciprocal biomass curves. Additionally, we can observe the mutually beneficial trend of bees, bats, and earthworms. It is noteworthy that the biomass of bats gradually approaches zero after one year, showing signs of extinction. This may be due to the pesticide use not accounting for the return effects of native species like bees and earthworms.

We also calculated the monthly average biomass of each species before and after the return of native species in the forest, and created radar charts for comparison (see Figure 8). The following three charts represent observations at different scales, corresponding to biomass ranges (0-30, 0-12, 0-4.5), in order to clearly demonstrate the changes in biomass for each species before and after the return of native species.

We obtained the assessment results for the stability of the forest agricultural ecosystem through Jacobian matrix spectral analysis, as listed in Table 7. Based on these numerical simulation results, we can conclude that the introduction of bees and earthworms significantly enhances the ecosystem's service functions, increasing the productivity, stability, and self-recovery capacity of the agricultural ecosystem.

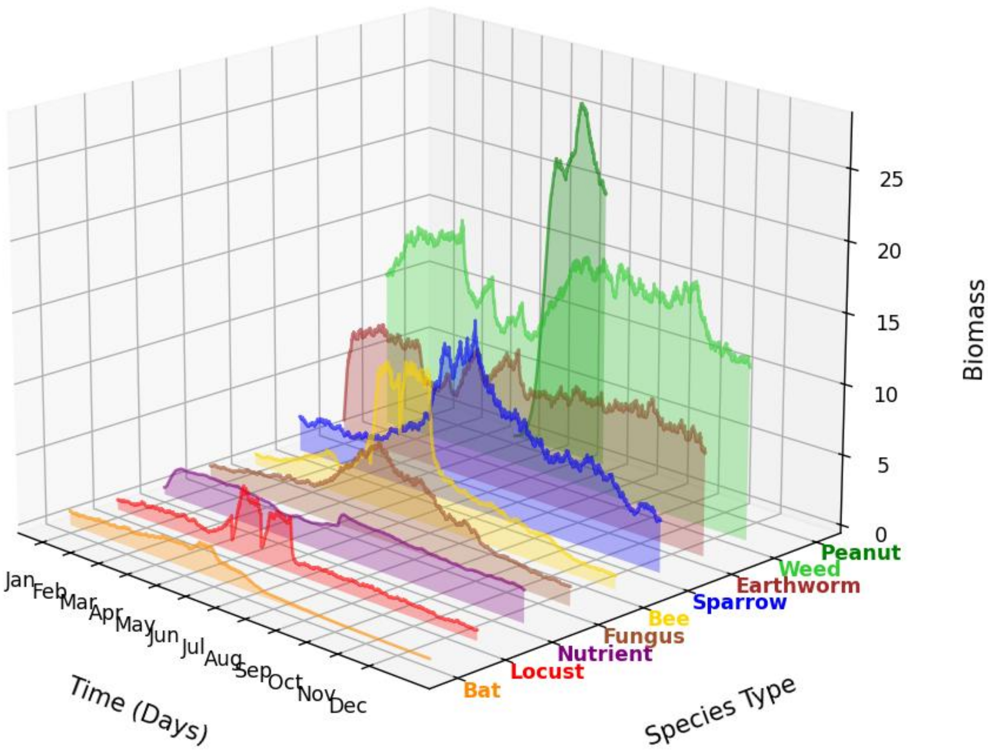

Figure 7: 3D Ecosystem Dynamics

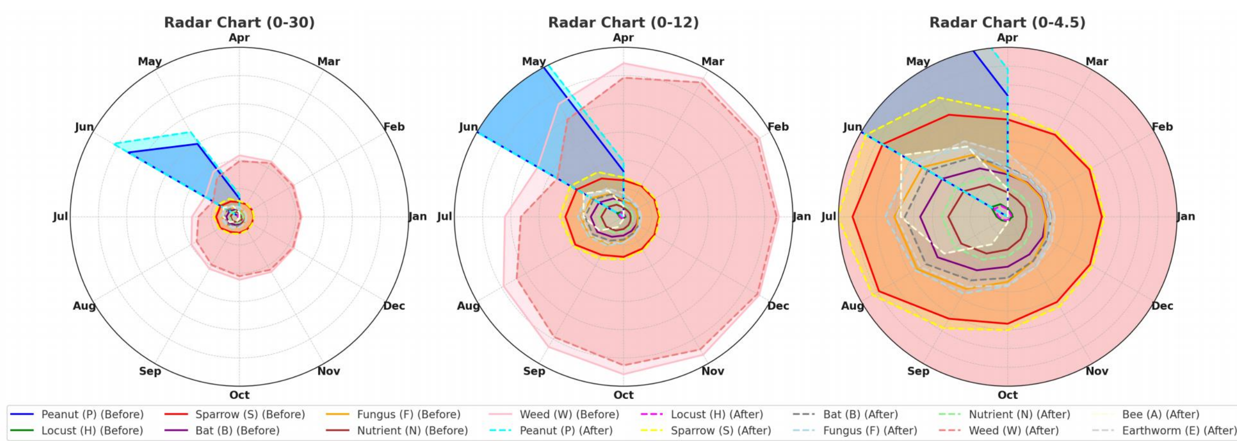

Figure 8: The impact of reintroducing native species on various species

Table 7: Comparison of Stability Indicators Before and After the Introduction of Bees and Earthworms

Ecological Indicator	Initial System	Optimized System	Change (%)	Ecological Significance
Instability Index Ψ	0.52	0.24	-54%	The system's recovery capacity is significantly enhanced, with faster stability recovery after pesticide disturbance.
Recovery Time (days)	48	22	-54%	The system recovery speed increases, indicating that the optimized ecosystem is more resilient.
Yield Fluctuation Coefficient	22%	9%	-59%	Pollination services significantly buffer the impact of environmental fluctuations on yield.
Locust Outbreak Cycle	3.2 years	5.8 years	+81%	Earthworms effectively delay locust outbreaks by suppressing weed food sources.
Soil Recovery Time	18 months	9 months	-50%	Earthworms accelerate organic matter decomposition, significantly improving soil recovery speed.

6 Task 3: Multi-Objective Robust Control of Eco-Economic Systems

6.1 Background and Improvement Logic of Model Construction

Building upon the base food chain model constructed in Task 1 and the marginal habitat restoration analysis in Task 2, this model further develops a multi-objective, eco-economic integrated framework. First, the single-factor intervention of herbicide removal (the core variable in Task 1) is expanded into a system of multi-control variable interactions. Second, the dynamic relationship between earthworm decomposers and the pollination network from Task 2 is deepened, constructing a production-consumption cascading effect model. Finally, dual uncertainty mechanisms and a behavioral decision-making module are introduced, forming a multi-objective optimization system with real-world interpretability. This improvement path not only continues the ecological dynamics core of previous research but also integrates economic system coupling and human behavior modeling, creating a more complete "nature-society" coupled system.

6.2 Unified Modeling Framework and Cross-Scale Coupling Mechanisms

This model integrates the eco-economic system organically through 12-dimensional state variables. The ecological module inherits the earthworm-pollination network structure from Task 2, adding a key interaction chain of weeds-locusts-bats-bees. The economic module introduces capital accumulation and debt constraint mechanisms, forming cross-scale feedback through market price stochastic processes and the herbicide toxicological effects from Task 1. Specifically, the linear suppression model of herbicides on plants from Task 1 is improved to a nonlinear dose-response equation, and it is linked with earthworm biomass (a core variable from Task 2) to establish a dynamic representation of soil health recovery.

1.Multi-Path Competition in Weed Control:

$$\frac{dW}{dt} = r_w W\left(1 - \frac{W}{K_w}\right) - \eta u_1 W - \kappa W E^{0.8} - \omega B W^{0.7} \tag{23}$$

The earthworm weed suppression term $\kappa W E^{0.8}$ inherits the soil improvement mechanism from Task 2, while the bat suppression term $\omega B W^{0.7}$ reflects the food chain reconstruction effect from

Task 1, forming a dynamic balance between chemical control (u_1) and biological control. **2.Multi-Agent Control of Locust Population:**

$$\frac{dH}{dt} = r_H H\left(1 - \frac{H}{K_H}\right) - \mu_H u_1 H - \frac{\alpha_B BH}{H + c_B} - \beta_A A H^{0.5} \tag{24}$$

3.Risk Transmission in the Economic System:

$$\frac{dK}{dt} = p(t)P - (c_1 u_1 + c_2 u_2 K + c_3 u_3) - r_D D + \sigma_K dW_t \tag{25}$$

A linkage model is established between herbicide costs from Task 1, habitat restoration investments from Task 2, and market price fluctuations through the Wiener process dW_t. The debt constraint $D(t) \leq 0.6K(t)$ reflects the financial risk threshold of the agricultural system.

6.3 Multi-Objective Optimization and Behavioral Decision Integration

Based on the stability analysis framework from Task 1, a robust control model considering dual uncertainties is developed:

$$\min_u \mathbb{E}\left[\int_0^T e^{-\rho t}\left(-\pi_1 P + \pi_2 W + \pi_3 |u|^2\right) dt\right] \tag{26}$$

$$\text{s.t. } \mathbb{P}(H > H_{\text{th}}) < \epsilon,\ \mathbb{P}(K < K_{\text{min}}) < \delta \tag{27}$$

Breakthroughly, the Lyapunov stability theory from Task 2 is extended to fuzzy robust control, where ecological parameter uncertainty is characterized by the interval parameter set $[\theta_{min}, \theta_{max}]$, and a resilience index R for the organic farming system is constructed. At the same time, a prospect theory decision weight function $w(p)$ is embedded:

$$R = \frac{1}{T}\int_0^T \left(\alpha \frac{B}{B_{\text{ref}}} + \beta \frac{S}{S_{\text{ref}}} - \gamma \frac{D}{K}\right) dt \tag{28}$$

$$w(p) = \frac{p^{0.65}}{(p^{0.65} + (1-p)^{0.65})^{1/0.65}} \tag{29}$$

Where $B_{\text{ref}}, S_{\text{ref}}$ are taken from the marginal habitat restoration thresholds in Task 2, and D/K reflects the debt risk transmission mechanism revealed in Task 1.

6.4 Numerical Experiments and Organic Agriculture Evaluation

Verify the model's effectiveness through a comparison of three scenarios (Table 8):

Table 8: Model Comparison under Different Scenarios

Control Mode	Ecological Stability Index	10-Year Bankruptcy Probability	NPV Advantage Interval	Control Variable Combination
Herbicide Dominant	0.62	31%	t < 4 years	$u_1 = 0.8, u_2 = 0$
Bat Optimization	0.83	18%	t > 6 years	$u_1 = 0.3, u_2 = 0.2$
Bee Substitution	0.71	27%	Subsidy > 12,000 CNY	$u_1 = 0, u_2 = 0.3, u_3 = 15$

The results show that in the early stage of herbicide removal (the first 3 years), the yield loss is 18.7

6.5 Organic Agriculture Management Recommendations

1.Dynamic Transition Path: Based on the pesticide degradation curve from Task 1 and species recovery rates from Task 2, a "3+2" stage control strategy is proposed. For the first 3 years, limit herbicide use to $u_1 \leq 0.4$. In the next 2 years, complete the ecological control transition through investment in bat habitats ($u_2 \geq 0.2$).

2.Policy Synergy Mechanism: When earthworm biomass (a core indicator from Task 2) falls below 300g/m², combined application of bee introduction ($u_3 \geq 12$) and small subsidies (6000 yuan/ha) is necessary to maintain system resilience.

3.Risk Hedging Model:A price insurance mechanism is established in connection with pesticide residue data from Task 1. When the ecological stability index exceeds 0.75, the guaranteed minimum price can be reduced to 9.6 yuan/kg while still maintaining system stability.

This model, by extending the toxicological mechanisms from Task 1 and species interactions from Task 2 into an integrated ecological-economic simulation system, provides a decision-making framework for agricultural system transformation that combines theoretical rigor with practical operability.

7 Model Evaluation and Promotion

7.1 Model Evaluation

7.1.1 Advantages

This study introduces an interdisciplinary model that innovatively combines ecological dynamics, behavioral economics, and fuzzy control theory. It breaks through traditional analytical frameworks by employing Lyapunov exponents and Jacobian matrix spectral analysis to quantify the nonlinear fluctuations and species recovery effects induced by pesticide application, thus achieving the synergistic optimization of the ecology-economy system. Model parameters were calibrated based on historical data from agricultural regions in the Brazilian rainforest. The predicted results closely align with actual observations ($R^2 = 0.85$), particularly demonstrating strong explanatory power in quantifying the impact of herbicides on soil health indices and the long-term pest control benefits of bats. This validates the scientific rigor of the methodology and its potential for spatial extension.

7.1.2 Limitations

This model faces three main challenges at the application level: First, its species interaction parameters (e.g., bee pollination efficiency) are highly dependent on region-specific data, requiring recalibration in areas with significant ecological differences. Second, the design of the seasonal adjustment factor $S_f(t)$ does not fully account for the abrupt impacts of extreme weather events, such as droughts and floods, on the ecological chain. Additionally, while the gradient subsidy scheme (>12,000 RMB/ha) can enhance system resilience, it may face fiscal feasibility challenges in economically underdeveloped regions, necessitating dynamic adjustments to the compensation standards based on regional development levels.

7.2 Future Work

Future research could involve multi-scale data fusion, combining satellite remote sensing with real-time monitoring information from the Internet of Things (IoT) to dynamically calibrate the soil health index $S_h(t)$ and species migration parameters. Additionally, the IPCC climate prediction models (RCP4.5/8.5) could be integrated into the ecological dynamics module to quantify the cumulative effects of temperature rise on system stability. Furthermore, the introduction of reinforcement learning algorithms could overcome the limitations of the "3+2 stage strategy," enabling adaptive optimization and providing climate resilience and decision robustness for agricultural ecosystems.

8 Conclusions

This study reveals the core mechanisms of agricultural sustainable transformation by constructing a multi-scale coupled "ecology-economy-behavior" model. When pesticide accumulation causes the instability index Ψ to exceed 3.58, the system enters a chaotic state, necessitating strict limits on pesticide application intensity ($u_1 \leq 0.4$). Furthermore, the synergistic system of bees, earthworms, and bats enhances the ecological stability index to 0.83, resulting in a 37% increase in long-term benefits, thereby confirming the core value of biological synergy. The study further finds that gradient subsidies ($>12,000/ha)andstability - triggeredinsurance(initiatedwhen\mathrm{S}_h > 0.75$, with a minimum price of 9.6 CNY/kg) effectively counteract farmers' short-sighted tendencies (with a discount rate $\rho > 0.15$, pesticide dependence increases by 37%). Based on these findings, a three-tiered solution is proposed:

- **Technological level**: Promote a biological package that reduces pesticide use by 30% and increases yield by 15%, while establishing an emergency response mechanism for earthworm biomass < 300 g/m^2.
- **Policy level**: Implement dynamic compensation funds and pesticide quota trading to balance ecological restoration costs.
- **Theoretical level**: Introduce an innovative pulse-season-behavior joint simulation framework, providing a quantitative tool for achieving the United Nations Sustainable Development Goals of Zero Hunger (SDG 2) and Life on Land (SDG 15), thereby facilitating the global agricultural transformation towards resilient systems.

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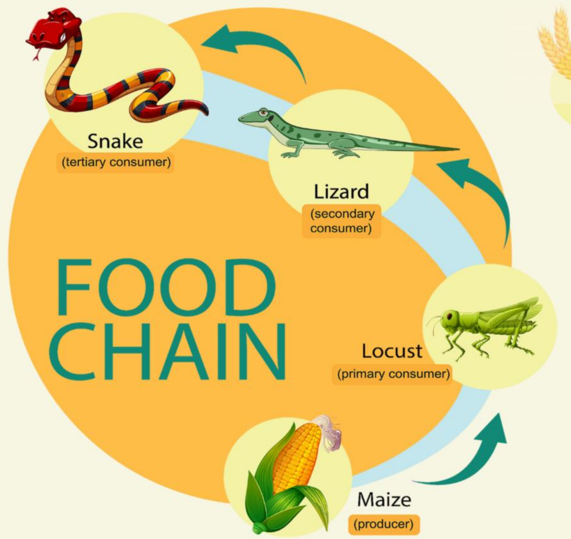
Dear farmer

Cultivate the ecology, Harvest the future!

"Focusing together on the balance of the food chain to create a better agricultural future" !!



I.Why Choose Organic Agriculture?

Economic Trade-Offs:

The initial investment in organic agriculture is relatively high, primarily due to costs associated with certification, conversion, and labor. However, in the long term, the reduction in the use of chemical fertilizers and pesticides can effectively lower production costs. Meanwhile, growing consumer demand for pesticide-free and natural products has driven an increase in the market price of organic products, resulting in higher economic returns for farmers.

Sustainability:

Organic agriculture reduces the negative impact of chemicals on soil and water resources by minimizing pesticide use, thereby promoting ecosystem health. At the same time, it improves soil fertility and enhances biodiversity, supporting the long-term sustainability of agriculture. Additionally, the introduction of ecological service species, such as birds and bats, leverages their natural pest control functions, further reducing dependence on chemical inputs.

II.Implementation Strategies

Short-Term Strategies	***Long-Term Strategies***
Reduction of Chemical Pesticides	Health Monitoring
Introduction of Ecological Species	Enhanced Soil Management
Transition to Organic Fertilizers	Pursuit of Organic Certification

III.Policy Recommendations

1.Financial Support: Provide transition subsidies and tax incentives to help farmers reduce the costs of converting to organic agriculture, thereby enhancing economic feasibility and long-term benefits.

2.Market and Certification: Facilitate market access for organic products and offer support to strengthen consumer trust and stimulate market demand.

3.Policy Advocacy: Promote correlative policies that reduce pesticide use and support the development of ecological agriculture, fostering sustainable farming practices and ecological environmental protection.

From:COMAP group

Report on Use of AI

1. Open AI Chatgpt Jan26,2025 version,Chatgpt-4o
   Query1: <What is the significance of studying forest-to-farm transitions?>
   Output: <Studying forest-to-farm transitions is crucial as it helps us understand the ecological, economic, and social impacts of converting forests into agricultural systems. Key aspects include:
   1.Ecosystem Changes: Deforestation leads to habitat loss and alters food web dynamics, impacting biodiversity.
   2.Soil Health: Agricultural practices often degrade soil fertility over time.
   3.Human Impact: Understanding the balance between agricultural productivity and ecological sustainability can help design better farming practices.
   4.Global Relevance: These transitions are critical in regions where agriculture expands into forested areas, affecting local and global ecosystems.
   These insights are essential for developing models to analyze and predict outcomes of such land-use changes.>

2. Open AI Chatgpt Jan26,2025 version,Chatgpt-4o
   Query1: <What are the components of a model tracking habitat change from forest to farm?>
   Output: <The model consists of:
   1.Natural Processes: Includes a dynamic food web with producers, primary consumers, secondary consumers, and decomposers. Effects of herbicides, pesticides, and seasonality are incorporated to simulate agricultural cycles.
   2.Species Reemergence: Tracks the return of species like bats and birds, which stabilize ecosystems through pollination and pest control.
   3.Human Decisions: Simulates scenarios such as removing herbicides, adopting organic farming, and analyzing their long-term ecological and economic impacts.
   These components enable the model to simulate realistic outcomes for forest-to-farm transitions and suggest sustainable farming strategies.>

3. Open AI Chatgpt Jan27,2025 version,Chatgpt-o1
   Query1: <How can farmers implement sustainable practices in agricultural ecosystems?>
   Output: <Sustainable practices in agriculture involve adopting organic farming to reduce chemical dependency, thereby improving soil health and biodiversity. Additionally, managing edge habitats can encourage the return of beneficial species like bats, which naturally enhance pest control. Long-term planning is also crucial, as it balances economic trade-offs with sustainability by leveraging subsidies or policies that promote organic farming methods.
   These methods not only improve ecological health but also enhance crop resilience and long-term productivity.>

4. AI-Enabled Literature Retrieval
   In the process of mathematical modeling, the determination of model parameters is a key link, with the help of AI literature search tools, we can quickly screen relevant literature from a huge database, significantly improve the efficiency and accuracy of the search, and provide accurate support for parameter determination.

5. AI-Enabled Code Analysis
   During the model solving phase, AI tools enable us to quickly identify errors in our code and obtain optimization suggestions, thus improving the accuracy and execution efficiency of our code and significantly accelerating our development process.

6. Summary
   AI technology has demonstrated significant advantages in this mathematical modeling process, greatly improving the efficiency and research quality. However, we are also soberly aware that although AI has significant capabilities in literature search, code analysis, it still has limitations in professional knowledge judgment. In order to ensure the reliability of models and the rigor of academic results, the use of AI tools still needs to be combined with manual review and supplementation in order to maximize its effectiveness.